

UNIVERSITÀ COMMERCIALE LUIGI BOCCONI

COMPUTER VISION AND IMAGE PROCESSING

Assignment 1



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1. Image Acquisition

The selected scene is the SDA Bocconi sign located on campus, as suggested in the assignment. Two images of a real 3D scene were acquired using the same camera device (iPhone 15), keeping focal length fixed (no digital zoom) and identical resolution. The two views were obtained by performing a small lateral translation of the camera while preserving approximately the same orientation. The scene contains multiple depth layers (foreground flowers, mid-ground sign, background buildings/sky), providing sufficient 3D structure for two-view geometry.



Fig. 1. The two views of the SDA's sign used.

2. Camera Calibration

We calibrate the camera using a planar checkerboard and Zhang's method. Figure 2 shows example distorted/undistorted pairs.

A. Camera model. The pinhole camera model is adopted. A 3D point in homogeneous coordinates $\mathbf{X} = (X, Y, Z, 1)^\top$ is projected to the image plane as

$$\lambda \mathbf{x} = \mathbf{K} [\mathbf{R} \mid \mathbf{t}] \mathbf{X}, \quad [1]$$

where $\mathbf{x} = (u, v, 1)^\top$ are homogeneous image coordinates, $\mathbf{R} \in SO(3)$ and $\mathbf{t} \in \mathbb{R}^3$ are the extrinsic parameters, and $\mathbf{K}$ is the intrinsic calibration matrix:

$$\mathbf{K} = \begin{bmatrix} f_x & s & c_x \\ 0 & f_y & c_y \\ 0 & 0 & 1 \end{bmatrix}. \quad [2]$$

B. Planar calibration via homographies: Zhang's method. The checkerboard is planar. By choosing the checkerboard reference frame such that $Z = 0$, each calibration point becomes $\mathbf{X}_{\text{plane}} = (X, Y, 1)^\top$. For each view j , the projection reduces to a planar homography:

$$\mathbf{x} \sim \mathbf{H}_j \mathbf{X}_{\text{plane}}. \quad [3]$$

Moreover, from [1] and $Z = 0$ one obtains

$$\mathbf{H}_j \sim \mathbf{K} [\mathbf{r}_{1j} \quad \mathbf{r}_{2j} \quad \mathbf{t}_j]. \quad [4]$$

Let $\mathbf{h}_{1j}$ and $\mathbf{h}_{2j}$ denote the first two columns of $\mathbf{H}_j$. Since $\mathbf{r}_{1j}$ and $\mathbf{r}_{2j}$ are orthonormal,

$$\mathbf{r}_{1j}^\top \mathbf{r}_{2j} = 0, \quad \|\mathbf{r}_{1j}\| = \|\mathbf{r}_{2j}\|. \quad [5]$$

Using [4], we write: $\mathbf{r}_{ij} = \lambda_j \mathbf{K}^{-1} \mathbf{h}_{ij}$ and define

$$\mathbf{B} = \mathbf{K}^{-\top} \mathbf{K}^{-1}. \quad [6]$$

Finally, [5] yields the linear constraints

$$\mathbf{h}_{1j}^\top \mathbf{B} \mathbf{h}_{2j} = 0, \quad \mathbf{h}_{1j}^\top \mathbf{B} \mathbf{h}_{1j} = \mathbf{h}_{2j}^\top \mathbf{B} \mathbf{h}_{2j}. \quad [7]$$

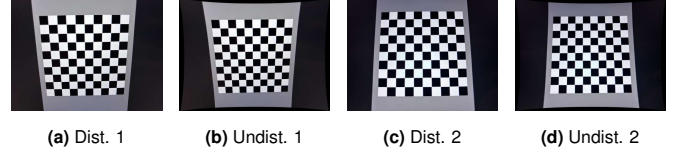


Fig. 2. Two of the seven pairs of checkerboard pictures used to calibrate.

Stacking [7] over multiple views gives a linear system in the entries of $\mathbf{B}$, solved in least squares sense (SVD). Finally, $\mathbf{K}$ is recovered from $\mathbf{B}$ (e.g., via Cholesky factorization), and extrinsics $(\mathbf{R}_j, \mathbf{t}_j)$ are computed for each view using [4].

C. Lens distortion model. Lens distortion is modeled as radial and tangential distortion. Let (x, y) be normalized coordinates:

$$x = \frac{X_c}{Z_c}, \quad y = \frac{Y_c}{Z_c}, \quad r^2 = x^2 + y^2. \quad [8]$$

A common radial+tangential model is

$$x_d = x (1 + k_1 r^2 + k_2 r^4 + k_3 r^6) + 2p_1 xy + p_2 (r^2 + 2x^2), \quad [9]$$

$$y_d = y (1 + k_1 r^2 + k_2 r^4 + k_3 r^6) + p_1 (r^2 + 2y^2) + 2p_2 xy. \quad [10]$$

OpenCV's `CALIB_RATIONAL_MODEL` was enabled in our implementation, which extends the radial model with additional higher-order terms (estimated by non-linear optimization together with other parameters).

D. Non-linear refinement. After the linear initialization (homographies + constraints), all parameters are refined by minimizing the reprojection error:

$$\min_{\mathbf{K}, \{\mathbf{R}_j, \mathbf{t}_j\}, \boldsymbol{\theta}} \sum_j \sum_i \|\mathbf{x}_{ij} - \pi(\mathbf{K}, \boldsymbol{\theta}, \mathbf{R}_j, \mathbf{t}_j, \mathbf{X}_i)\|^2, \quad [11]$$

where $\pi(\cdot)$ denotes the full projection function including distortion. OpenCV's Levenberg-Marquardt refinement was used.

E. Calibration setup and results.

Calibration setup: $N = 7$ checkerboard images were used.

Intrinsic matrix: The estimated intrinsic matrix is:

$$\mathbf{K} = \begin{bmatrix} 4546.25 & 0 & 2864.75 \\ 0 & 4545.04 & 2251.93 \\ 0 & 0 & 1 \end{bmatrix}. \quad [12]$$

Distortion coefficients: With OpenCV rational model (`CALIB_RATIONAL_MODEL`) enabled, the estimated parameters are:

$$\boldsymbol{\theta} = \begin{bmatrix} k_1 \\ k_2 \\ p_1 \\ p_2 \\ k_3 \\ k_4 \\ k_5 \\ k_6 \end{bmatrix} = \begin{bmatrix} 163.85 \\ 72.67 \\ 5.65 \times 10^{-4} \\ 3.95 \times 10^{-4} \\ -31.81 \\ 152.78 \\ 79.13 \\ 1.29 \end{bmatrix}. \quad [13]$$

The remaining six higher-order coefficients are zero.

Reprojection error: The mean reprojection error across all views is reported as the average L_2 distance between detected

Image	Reprojection error [px]
IMG_6971.jpg	~0.83
IMG_6972.jpg	~0.83
IMG_6973.jpg	~0.83
IMG_6974.jpg	~0.83
IMG_6975.jpg	~0.83
IMG_6976.jpg	~0.83
IMG_6977.jpg	~0.83
Overall mean	0.826

Table 1. Per-view reprojection errors returned by OpenCV (mean L_2 distance). Overall mean: 0.826 px.

and reprojected corners. The overall mean reprojection error is **0.826** px, which confirms a reliable calibration.

Discussion: It can be observed that $f_x \approx f_y$ (nearly square pixels), skew close to zero, and a principal point close to the image center. The low reprojection error indicates an accurate calibration.

3. Feature Correspondences

Corresponding points in the two images were annotated:

$$\{(\mathbf{x}_i, \mathbf{x}'_i)\}_{i=1}^N, \quad \mathbf{x}_i = (u_i, v_i, 1)^\top, \quad \mathbf{x}'_i = (u'_i, v'_i, 1)^\top.$$



Fig. 3. SIFT and manual correspondences between images.

A total of $N = 438$ correspondences was used: 220 manually annotated points on the SDA Bocconi sign letters and surrounding landmarks (trees, bench, flower, pole cover), plus 218 automatically extracted SIFT correspondences on the ground and surrounding areas.

For the automatic pipeline, a SIFT grid search was run across multiple parameter configurations (number of features, contrast threshold, edge threshold, Lowe's ratio, RANSAC threshold, and matcher type), selecting the configuration that maximized an inlier-count/Sampson-error score. The SIFT inliers were then filtered by Sampson distance (< 0.5), spatially thinned (minimum nearest-neighbour distance of 12 px), and de-duplicated against manual annotations (minimum distance of 18 px) before merging.

4. Epipolar Geometry and Two-View Reconstruction

A. From pixel coordinates to calibrated coordinates. Using the estimated intrinsics $\mathbf{K}$, pixel coordinates are mapped to normalized camera coordinates:

$$\mathbf{p}_i = \mathbf{K}^{-1} \mathbf{x}_i, \quad \mathbf{q}_i = \mathbf{K}^{-1} \mathbf{x}'_i,$$

so that the epipolar constraint is expressed with the essential matrix $\mathbf{E}$:

$$\mathbf{q}_i^\top \mathbf{E} \mathbf{p}_i = 0. \quad [14]$$



Fig. 4. Display of epipolar lines.

B. Normalized Eight-Point Algorithm for $\mathbf{E}$. Given $N \geq 8$ correspondences in normalized coordinates $\{(\mathbf{p}_i, \mathbf{q}_i)\}_{i=1}^N$, $\mathbf{E}$ is estimated by solving a linear system and enforcing the essential constraints.

(1) Hartley normalization: A similarity transform is applied to improve numerical conditioning:

$$\hat{\mathbf{p}}_i = \mathbf{T} \mathbf{p}_i, \quad \hat{\mathbf{q}}_i = \mathbf{T}' \mathbf{q}_i,$$

where $\mathbf{T}, \mathbf{T}'$ translate points to zero-mean and scale them so that the mean distance to the origin is $\sqrt{2}$.

(2) Linear system construction: Each correspondence yields one row of $\mathbf{A} \mathbf{e} = \mathbf{0}$.

(3) Solve by SVD and enforce essential constraints: The solution for $\hat{\mathbf{E}}$ is obtained via SVD and the singular value structure is enforced:

$$\hat{\mathbf{E}} = \mathbf{U} \text{diag}(\sigma_1, \sigma_2, \sigma_3) \mathbf{V}^\top \Rightarrow \hat{\mathbf{E}}_{\text{ess}} = \mathbf{U} \text{diag}(1, 1, 0) \mathbf{V}^\top.$$

$$\mathbf{E} = \mathbf{T}'^\top \hat{\mathbf{E}}_{\text{ess}} \mathbf{T}. \quad [15]$$

C. Estimated essential matrix. The fundamental matrix estimated by the eight-point algorithm is:

$$\mathbf{F} = \begin{bmatrix} -1.480 \times 10^{-8} & 5.178 \times 10^{-7} & -1.114 \times 10^{-4} \\ -3.305 \times 10^{-7} & 1.224 \times 10^{-7} & -3.340 \times 10^{-3} \\ 8.058 \times 10^{-5} & 2.459 \times 10^{-3} & 1 \end{bmatrix}. \quad [16]$$

From this, the essential matrix $\mathbf{E} = \mathbf{K}^\top \mathbf{F} \mathbf{K}$ is:

$$\mathbf{E} = \begin{bmatrix} -0.306 & 10.700 & 4.602 \\ -6.829 & 2.528 & -18.230 \\ -3.210 & 19.171 & 0.636 \end{bmatrix}. \quad [17]$$

D. Epipolar lines. For visualization in pixel coordinates, $\mathbf{E}$ is converted to the fundamental matrix:

$$\mathbf{F} = \mathbf{K}^{-\top} \mathbf{E} \mathbf{K}^{-1}.$$

For each point $\mathbf{x}'_i$ in image 2, the corresponding epipolar line in image 1 is

$$\ell_i = \mathbf{F}^\top \mathbf{x}'_i.$$

The epipolar lines are plotted in both images. For each point in image 2, the corresponding epipolar line in image 1 is drawn in blue; for each point in image 1, the line in image 2 is drawn in red.

E. Error metrics. The quality of $\mathbf{E}$ (or $\mathbf{F}$) is evaluated using the Sampson distance:

$$d_{\text{Sampson}}(\mathbf{x}_i, \mathbf{x}'_i) = \frac{(\mathbf{x}'_i{}^\top \mathbf{F} \mathbf{x}_i)^2}{(\mathbf{F} \mathbf{x}_i)_1^2 + (\mathbf{F} \mathbf{x}_i)_2^2 + (\mathbf{F}^\top \mathbf{x}'_i)_1^2 + (\mathbf{F}^\top \mathbf{x}'_i)_2^2}.$$

Metric	Value [px ²]
Mean Sampson distance	21.98
Median Sampson distance	0.62
Max Sampson distance	2182.52

Table 2. Geometric consistency errors for the estimated epipolar geometry (custom eight-point, all 309 points).

Method	# Inliers	Mean Sampson (inliers) [px ²]
Eight-point (ours)	438/438	21.98
Five-point (OpenCV + RANSAC)	167/438	1.61

Table 3. Comparison between our eight-point estimation and OpenCV five-point estimation.

F. Optional non-linear refinement. No explicit non-linear refinement of $\mathbf{E}$ was performed beyond the linear eight-point estimate. However, OpenCV’s five-point algorithm internally uses RANSAC, which removes outliers and yields a cleaner estimate.

G. Comparison with OpenCV five-point algorithm. The eight-point estimate was compared with OpenCV’s five-point algorithm for $\mathbf{E}$. The function `cv2.findEssentialMat` was called with `method=cv2.RANSAC`, `prob=0.999`, and `threshold=1.0` px.

Discussion: The five-point algorithm with RANSAC achieves a drastically lower Sampson error because it rejects outliers and enforces the essential constraints in calibrated coordinates.

5. 3D Reconstruction

A. Pose recovery from $\mathbf{E}$. The essential matrix was decomposed as $\mathbf{E} = [\mathbf{t}]_{\times}\mathbf{R}$, yielding four candidate solutions; the physically valid one was selected via chirality. Using `cv2.recoverPose` on the OpenCV five-point $\mathbf{E}$:

$$\mathbf{R} = \begin{bmatrix} 0.9820 & -0.0765 & 0.1726 \\ 0.0624 & 0.9944 & 0.0854 \\ -0.1782 & -0.0731 & 0.9813 \end{bmatrix}, \quad \mathbf{t} = \begin{bmatrix} -0.845 \\ -0.196 \\ 0.497 \end{bmatrix}. \quad [18]$$

Note that $\mathbf{t}$ is a unit vector (scale ambiguity).

B. Triangulation. Given $\mathbf{P}_1 = [\mathbf{I}|\mathbf{0}]$ and $\mathbf{P}_2 = [\mathbf{R}|\mathbf{t}]$, points were triangulated using `cv2.triangulatePoints`. All 438 correspondences passed the chirality check. The mean reprojection error after triangulation is 3.18 px (camera 1) and 3.02 px (camera 2).

C. 3D visualization. The reconstructed 3D point cloud is visualized from multiple viewpoints. The notebook (`ex2hwmanual.ipynb`) generates static views and a 360° animation (GIF). An additional interactive Plotly scatter guarantees a complete view of the reconstructed scene.

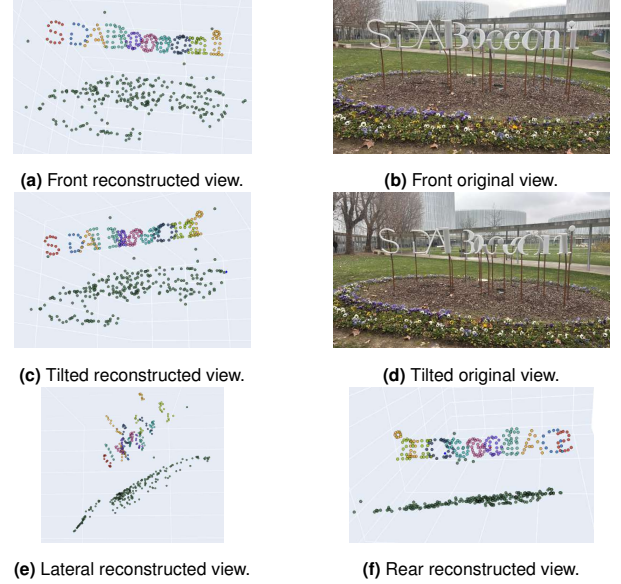


Fig. 5. 3×2 comparison between reconstructed 3D viewpoints and the original images, including additional perspectives (lateral and rear).

6. Conclusion

A full two-view reconstruction pipeline was completed, which included: calibration (mean reprojection error 0.83 px), epipolar geometry estimation (custom eight-point vs. OpenCV five-point with RANSAC) and sparse 3D triangulation. Remaining limitations include scale ambiguity and sensitivity to correspondence noise.

Beyond the quantitative results, this project highlights several key geometric aspects of two-view vision:

First, the reconstruction obtained from point correspondences is inherently defined only up to a global scale factor, reflecting the well-known projective ambiguity of structure from motion. The translation vector recovered from the essential matrix is therefore unit-norm and encodes only direction. This intrinsic ambiguity is not a limitation of the implementation, but a fundamental property of perspective imaging.

Second, the comparison between the normalized eight-point algorithm and OpenCV’s five-point method clearly illustrates the importance of robust estimation. Although the eight-point algorithm provides a valid linear solution using all correspondences, its sensitivity to outliers results in a significantly higher mean Sampson error. On the other hand, the five-point algorithm combined with RANSAC effectively rejects mismatches and enforces the essential matrix constraints during estimation, leading to a substantially cleaner epipolar geometry. This confirms the theoretical expectation that robust consensus based methods are crucial in practical vision pipelines.

From a geometric standpoint, the visualization of the epipolar lines confirms the correctness of the estimated fundamental matrix, and the chirality check during pose recovery ensures that the physically valid camera configuration is selected among the four possible decompositions of the

essential matrix. The fact that 308 out of 309 triangulated points satisfy the positive depth constraint further validates the consistency of the recovered motion.

Although the resulting 3D reconstruction is sparse and noisy due to correspondence inaccuracies, limited baseline, and the absence of global bundle adjustment, it successfully captures the relative spatial layout of the scene. In particular, the separation between foreground, mid-ground, and background elements is clearly observable in the reconstructed point cloud, demonstrating the effectiveness of the two-view geometry framework.

In conclusion, our work demonstrates how classical multiple-view geometry, starting from camera calibration and progressing through epipolar constraints to triangulation, provides a complete and coherent mathematical framework for recovering 3D information from images. Despite the simplicity of the setup (two images acquired with a smartphone), the pipeline rebuilds the geometry of the scene in a clear and grounded way, effectively connecting core theory with real-world computer vision implementation.

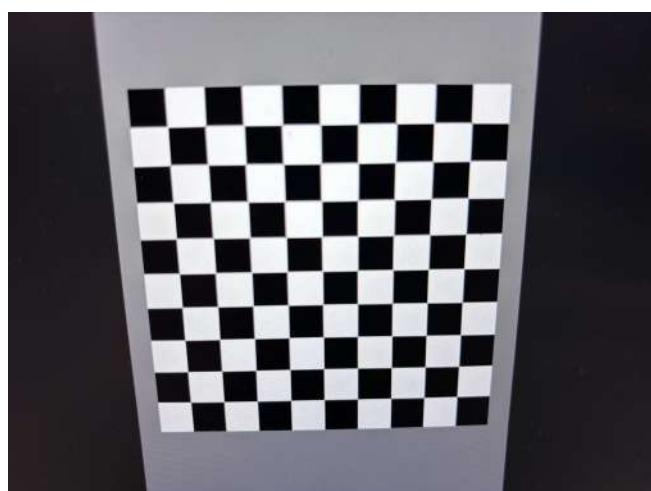
Annex: Large Figures



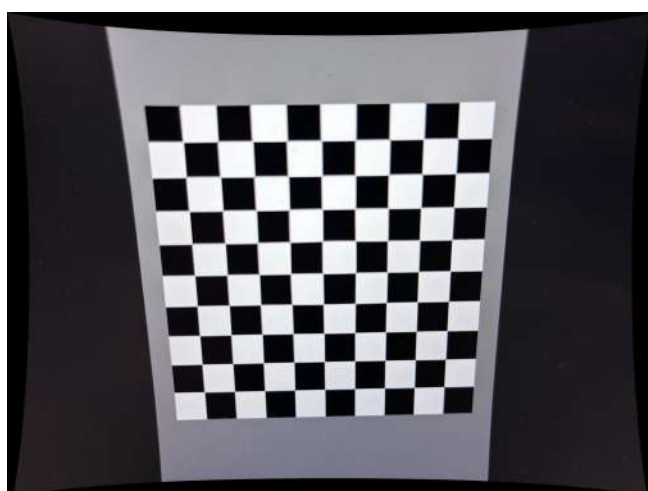
(a) View 1



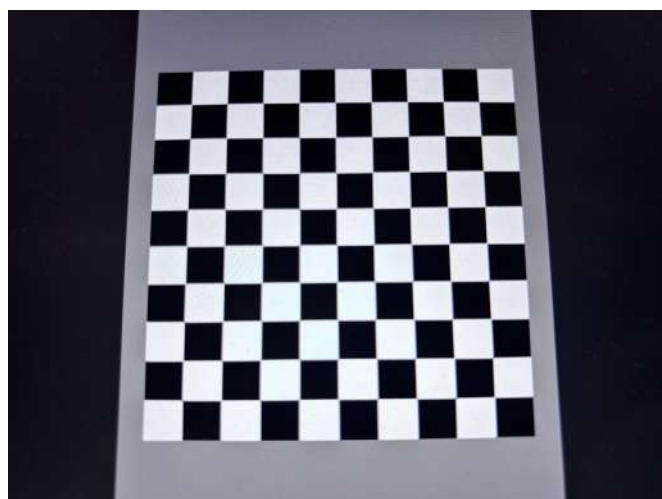
(b) View 2



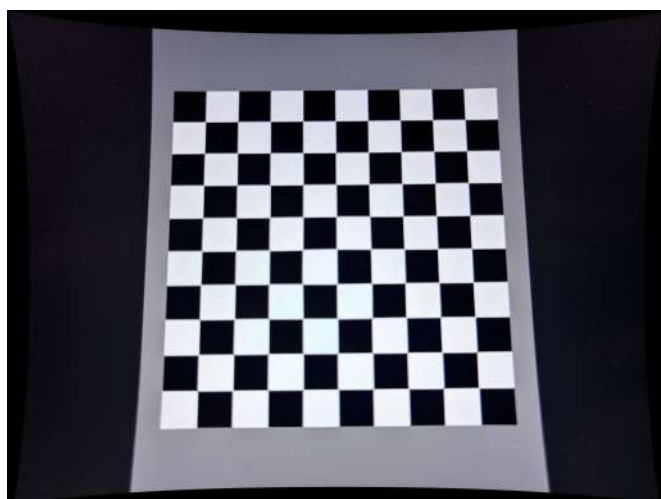
(c) Dist. 1



(d) Undist. 1



(e) [Dist. 2]



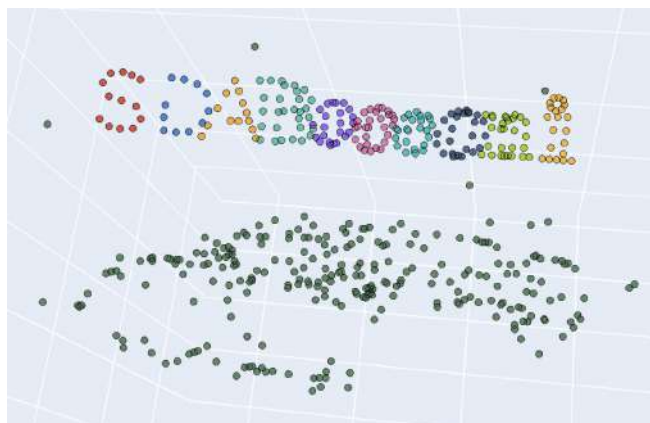
(f) [Undist. 2]



(a) SIFT and manual correspondences between images.



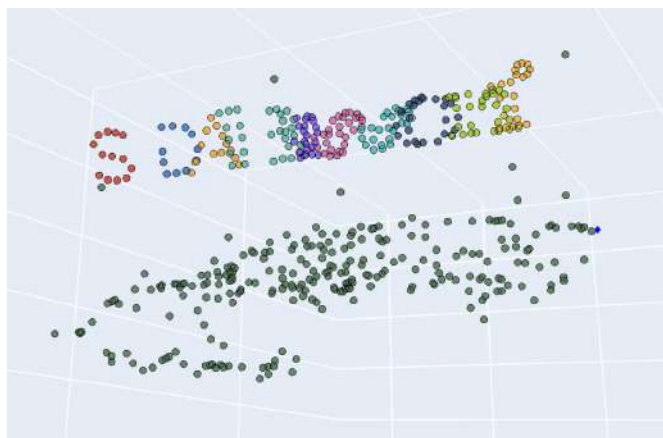
(b) Display of epipolar lines.



(c) Front reconstructed view.



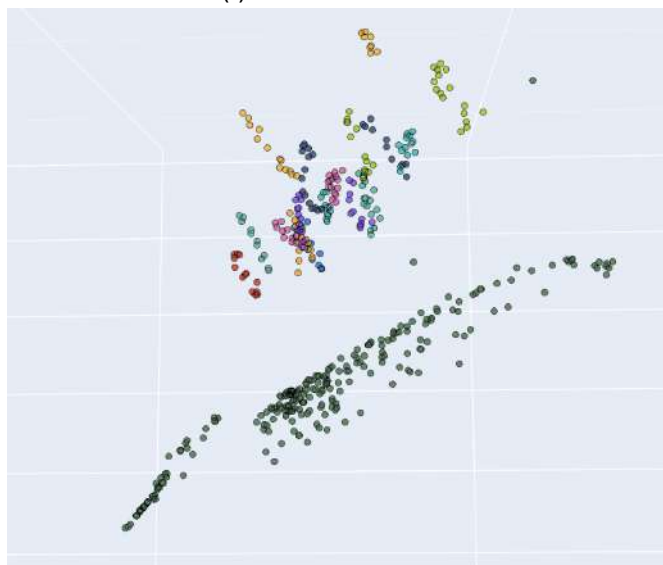
(d) Front original view.



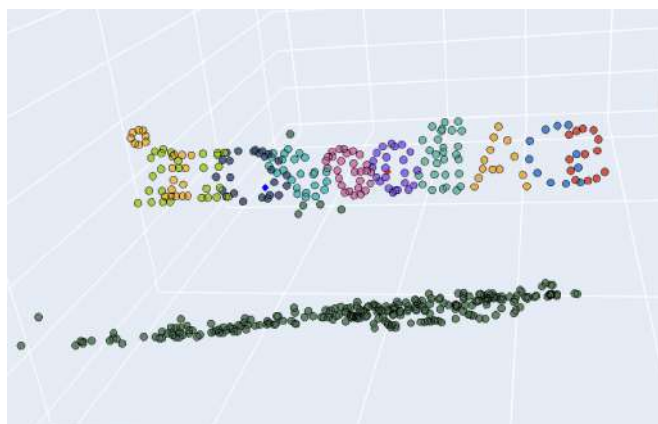
(e) Tilted reconstructed view.



(f) Tilted original view.



(g) Lateral reconstructed view.



(h) Rear reconstructed view.

References

- [1] G. Boracchi and L. Magri, *Computer Vision and Image Processing: Lecture Notes*, Università Commerciale Luigi Bocconi, Milano, 2026.
- [2] R. I. Hartley and A. Zisserman, *Multiple View Geometry in Computer Vision*, 2nd ed., Cambridge University Press, 2004.
- [3] Anthropic, *Claude (Version 4.6)*, Large Language Model (LLM), accessed March 2026. Available at: <https://claude.ai/new>